ISLAMIC UNIVERSITY OF TECHNOLOGY

Course Outline and Course Plan

Name of the Teacher	Ishmam Tashdeed	**Position**	Lecturer
Department	CSE	**Program**	B.Sc. Engineering in SWE
Course Code	SWE 4537	**Course Title**	Server Programming
Academic Year	2025-26	**Semester**	Winter
Contact Hours	3.00	**Credit Hours**	3.00
Prerequisites	A firm grasp of software architecture and development, Database Fundamentals	**Curriculum Requirement**	Compulsory
Course Homepage	Google Classroom Code: bodfypyd		
Teaching Methods/ Approaches	Lecture Demonstration		
Teaching Aids	Multimedia and OHP Board and Marker		

Course Assessment Method

Attendance (10%)	**Quiz/Viva (30%) (Best 3 out of 4)**				**Mid Exam (25%)**	**Final Exam (50%)**
Throughout the Semester	**1st Quiz**	**2nd Quiz**	**3rd Quiz**	**4th Quiz**	Week/Date	Week/Date
	Week/ Date	Week/ Date	Week/ Date	Week/ Date		
	4th Week	7th Week	11th Week	15th Week	8th Week	16th Week

Course Contents and Objectives	**Contents** Web service, HTTP protocols, IP, port, URL, routing. Web security fundamentals, CORS, authorization, authentication, OAuth, Social authentication, SSO. Database Connectivity, ORM. State Management, Session, Cookie, WebSocket, Server Push. Performance, scaling, load balancing, lazy loading, caching; Fundamentals of web service deployment, Background service, security and firewall, multi-threading.

	Objectives To theoretically analyze the structural constraints of REST and MVC architectures, evaluate the stateless mechanics of HTTP/HTTPS protocols, and examine the abstraction of the Object-Relational Impedance Mismatch in data persistence. Furthermore, the objective is to critique identity management frameworks by contrasting stateful and stateless JWT models while investigating the distributed systems principles underlying scalability and persistent WebSocket communication.
Course Outcomes	**CO1** — Explain the fundamental concepts of server-side communication, including the HTTP lifecycle, statelessness, RESTful architectural styles, and state management mechanisms. **CO2** — Design conceptual server-side architectures that integrate authentication flows, database interactions, and middleware strategies to meet functional requirements. **CO3** — Analyze system performance, bottlenecks, and formulate optimization strategies using caching mechanisms, load balancing, and horizontal scaling concepts. **CO4** — Evaluate and select different deployment environments or tools and security best practices to ensure application reliability, scalability, and data integrity in production systems.

Weekly Plan for Course Content		
Weeks	**Topics**	**Task/Reading**
1	Foundation of Web: Lec 1: What is a Server? (IP, Port, URL, HTTP protocols) Lec 2: What is an API?(Web Services, REST, Architectural Pattern/MVC)	Lecture Material and Class notes
2	Building the API: Lec 3: Intro to the Stack (Python/FastAPI) & Routing Lec 4: Database Connectivity & ORM	Lecture Material and Class notes
3	Authentication: Lec 5: State Management (Session, Cookie), AuthN Vs. AuthZ Lec 6: Stateless Auth (Token-based auth (JWTs) & Web security fundamentals.	Lecture Material and Class notes
4	Browser Security Lec 7: Browser Security (CORS) Lec 8: Practical API Security (Refining the routes)	Lecture Material and Class notes
5	Third-Party Authentication Lec 9: OAuth, Social authentication, SSO	Lecture Material and

	Lec 10: OAuth (Practical code flow)	Class notes
6	Advanced Connections: Lec 11: WebSocket & Server Push Lec 12: Diffie-Hellman key exchange	Lecture Material and Class notes
7	Performance & Caching Lec 13: Performance & Caching (Intro to Redis) Lec 14: Mid Semester Exam Review	Lecture Material and Class notes
8	Practical Caching and Async Jobs Lec 15: Caching Patterns (Cache-Aside, Write-Through) Lec 16: Background Service (Job Queues e.g. Celery)	Lecture Material and Class notes
9	Scaling & Architecture: Lec 17: Horizontal Scaling & Load Balancing (Nginx) Lec 18: Database Scaling (Replication vs. Sharding)	Lecture Material and Class notes
10	Advanced Data Distribution Lec 19: CAP Theorem & Distributed Consensus	Lecture Material and Class notes
11	Deployment & Production Lec 20: Containerization (Docker) & Orchestration concepts, Fundamental of deployment	Lecture Material and Class notes
12	Data-Intensive Systems Lec 21: Search Engines (ElasticSearch)	Lecture Material and Class notes
13	Lec 22: Security & Firewalls (WAF)	Lecture Material and Class notes
14	Lec 23: Batch Processing (MapReduce/Hadoop) Lec 24: Semester Final Review	Lecture Material and Class notes

Mapping of Course Outcomes (COs) and Program Outcomes (POs) and Evaluation Methods				
Assessment Method	**Marks**	**Mark Distributions (as %) on COs and POs**		
		CO1	**CO2**	**CO3**
		PO1	**PO2**	**PO2**
Attendance (Class Participation)	10%			
Quiz 1/Quiz 2/Quiz 3/Quiz 4	30%			
Mid Semester Exam	25%			

Final Exam	50%			
Total	100%			

Mapping of COs and POs												
Course Outcomes	**Program Outcomes (POs)**											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
CO1	✓											
CO2			✓									
CO3		✓										
CO4					✓							

Program Outcomes (POs)	
Students graduating from the Bachelor of Science in Computer Science and Engineering (B. Sc. in CSE) program, upon graduation, will have the ability to:	
PO 1	**Engineering Knowledge** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO 2	**Problem Analysis** Identify, formulate, research, and analyze complex engineering problems and reach substantiated conclusions using the principles of mathematics, the natural sciences, and the engineering sciences.
PO 3	**Design/Development of Solutions** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety and of cultural, societal, and environmental concerns.
PO 4	**Investigation** Conduct investigations of complex problems, considering experimental design, data analysis, and interpretation, and information synthesis to provide valid conclusions.
PO 5	**Modern Tool Usage** Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering activities with an understanding of their limitations.
PO 6	**The Engineer and Society** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional

	engineering practice.
PO 7	**Environment and Sustainability** Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of and need for sustainable development.
PO 8	**Ethics** Apply ethical principles and commit to the professional ethics, responsibilities and norms of the engineering practice.
PO 9	**Individual Work and Teamwork** Function effectively as an individual and as a member or leader of diverse teams and in multidisciplinary settings.
PO 10	**Communication** Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.
PO 11	**Project Management and Finance** Demonstrate knowledge and understanding of engineering and management principles and apply these to one's work as a team member or a leader to manage projects in multidisciplinary environments.
PO 12	**Life-Long Learning** Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.

K

Table: Knowledge Profile	
Attribute	
K1	A systematic, theory-based understanding of the natural sciences applicable to the discipline
K2	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline
K3	A systemic, theory-based formulation of engineering fundamentals required in the engineering discipline
K4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the

	forefront of the discipline
K5	Knowledge that supports engineering design in a practice area
K6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
K7	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability
K8	Engagement with selected knowledge in the research literature of the discipline

P

Table: Range of Complex Engineering Problem Solving	
Attribute	**Complex Engineering Problems have characteristic P1 and some or all of P2 to P7**
Depth of knowledge required	P1: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6, or K8 which allows a fundamentals-based, first principles analytical approach
Range of conflicting requirements	P2: Involve wide-ranging or conflicting technical, engineering and other issues
Depth of analysis required	P3: Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models
Familiarity of issues	P4: Involve infrequently encountered issues
Extent of applicable codes	P5: Are outside problems encompassed by standards and codes of practice for professional engineering
Extent of stakeholder involvement and conflicting requirements	P6: Involve diverse groups of stakeholders with widely varying needs.
Interdependence	P7: Are high-level problems including many component parts or sub-problems

A

Table: Range of Complex Engineering Activities	
Attribute	**Complex activities means (engineering) activities or projects that have some or all of the following characteristics:**
Range of resources	A1: Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies)
Level of interaction	A2: Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues
Innovation	A3: Involve creative use of engineering principles and research-based knowledge in novel ways
Consequences for society and the environment	A4: Involve creative use of engineering principles and research-based knowledge in novel ways
Familiarity	A5: Can extend beyond previous experiences by applying principles-based approaches

Grading Policy		
Numeric Grade	**Letter Grade**	**Grade Point**
80% and above	A+	4.00
75% to less than 80%	A	3.75
70% to less than 75%	A-	3.50
65% to less than 70%	B+	3.25
60% to less than 65%	B	3.00
55% to less than 60%	B-	2.75
50% to less than 55%	C+	2.50
45% to less than 50%	C	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00

Class Schedule	
Day	**Time**
Monday	9:15 a.m.
Thursday	2:30 p.m.

Student Consulting Hour:

- Thursday, 9:30 am - 12:00 pm

Instruction Contact Details:

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